Projective genus-2 law follow-up

Frozen 21-model, 1,260-target paired cohort per arm

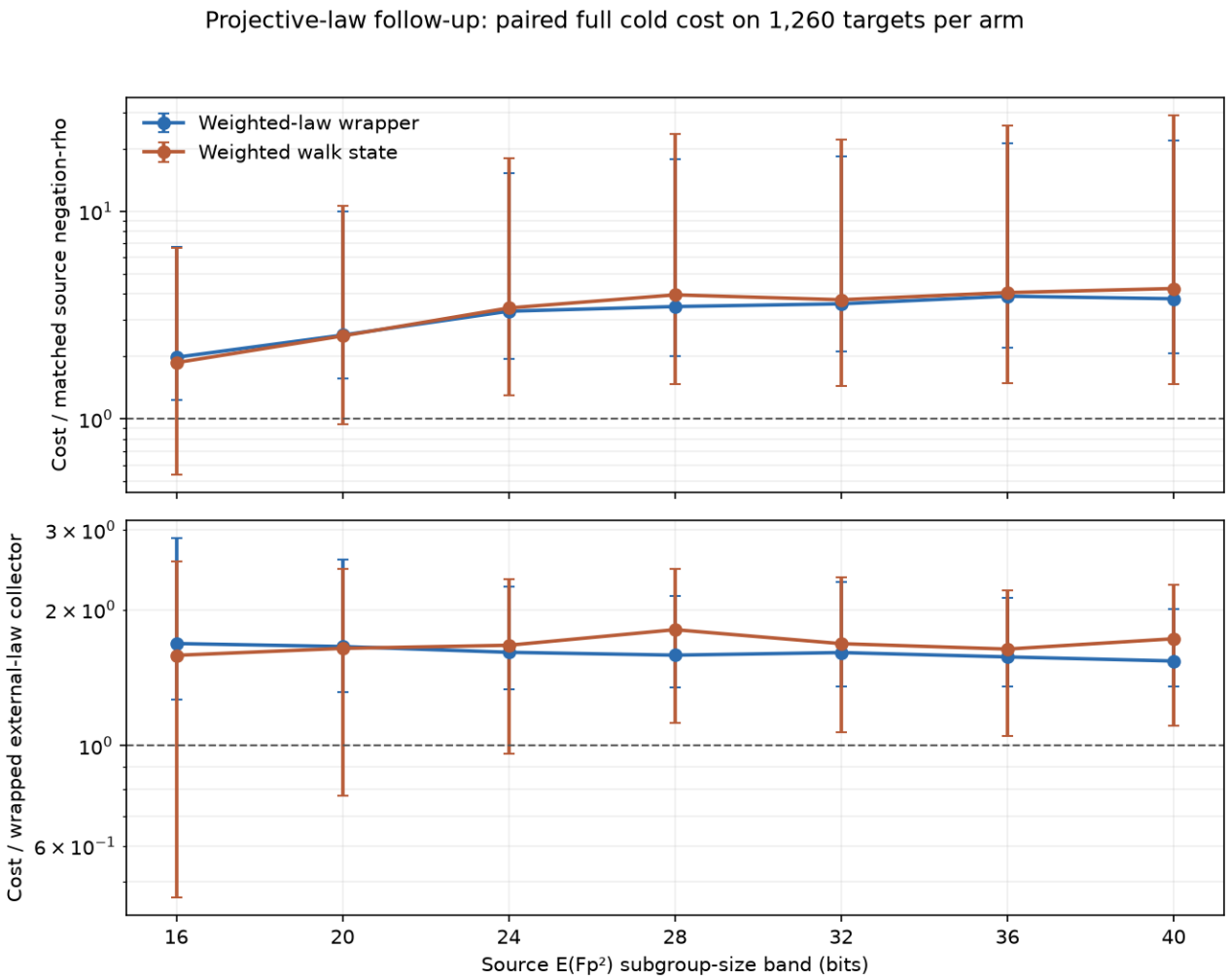
Scope. Two complete cold graph-collector arms use the independently checked weighted genus-2 law. The wrapper normalizes every projective output. The state arm keeps weighted divisors across walk steps and normalizes split candidates at the relation boundary. Each arm has 1,260 source-scalar-verified candidate solves and 1,260 same-process, independently source-verified negation-rho solves. The archived wrapped external-law collector is the second comparator. All ratios below divide the summed candidate cost by the summed comparator cost for the same targets.

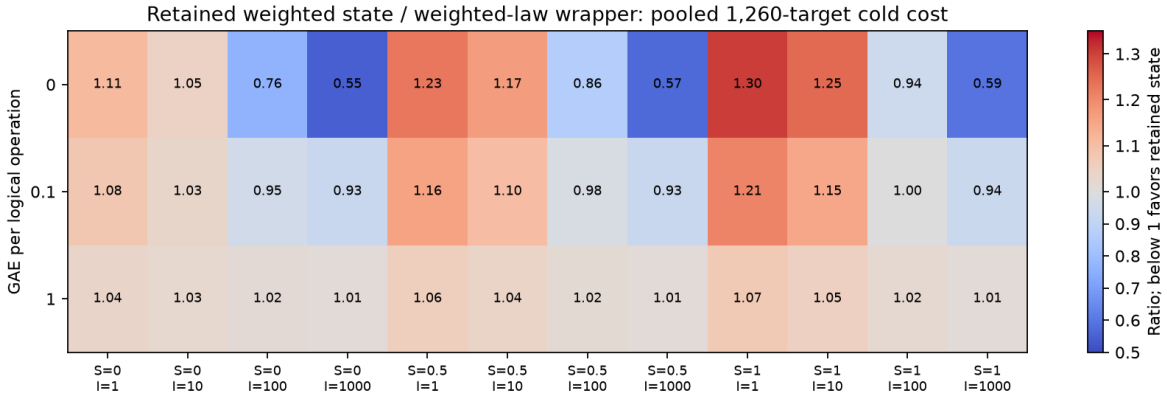
Bits	Wrapper/rho	State/rho	State/wrapper	Wrapper/external	State/external	State boundary I
16	1.9818	1.8665	0.9418	1.6825	1.5846	52,444
20	2.5346	2.5111	0.9907	1.6571	1.6418	208,610
24	3.3030	3.4237	1.0365	1.6092	1.6680	672,048
28	3.4777	3.9598	1.1386	1.5860	1.8059	2,811,932
32	3.5856	3.7514	1.0462	1.6071	1.6815	10,275,302
36	3.9013	4.0603	1.0407	1.5719	1.6359	41,087,298
40	3.7933	4.2482	1.1199	1.5395	1.7241	167,267,822
All	3.7938	4.1793	1.1016	1.5492	1.7067	222,375,456

Unit: source-addition equivalent (GAE). Representative price: $M + 0.5S + 10I$ plus 0.1 GAE per native logical operation. Boundary inversion counts are field inversions charged in the state arm; the wrapper's normalization inversions are also charged. Every ratio uses full cold setup, transfer, initialization, relation collection, elimination, and source verification. Pooled 36-cell wrapper/rho range: 2.0975–21.4887; state/rho range: 1.4729–28.0153. Cells below rho: wrapper 0/36; state 0/36. Pooled state/wrapper range: 0.5468–1.3037; 12/36 prices favor state and 24/36 favor wrapper. **Cold phases.** The standardized cold-phase-breakdown.csv reports arithmetic counts and source-addition-equivalent costs for construction, transfer, walk setup, relation collection and signed elimination, source verification, and a separate row for native logical work. Its phase rows sum to the total representative cost for each arm and comparator, for each bit size and all 1,260 targets.

40-bit method	Setup	Search/elimination	Native logical	Verification
Wrapper	1.206	2861.016	248.087	0.079
Weighted state	0.961	3228.833	253.458	0.079
External	0.669	1771.581	248.087	0.079
Rho	0.860	734.608	84.414	0.079

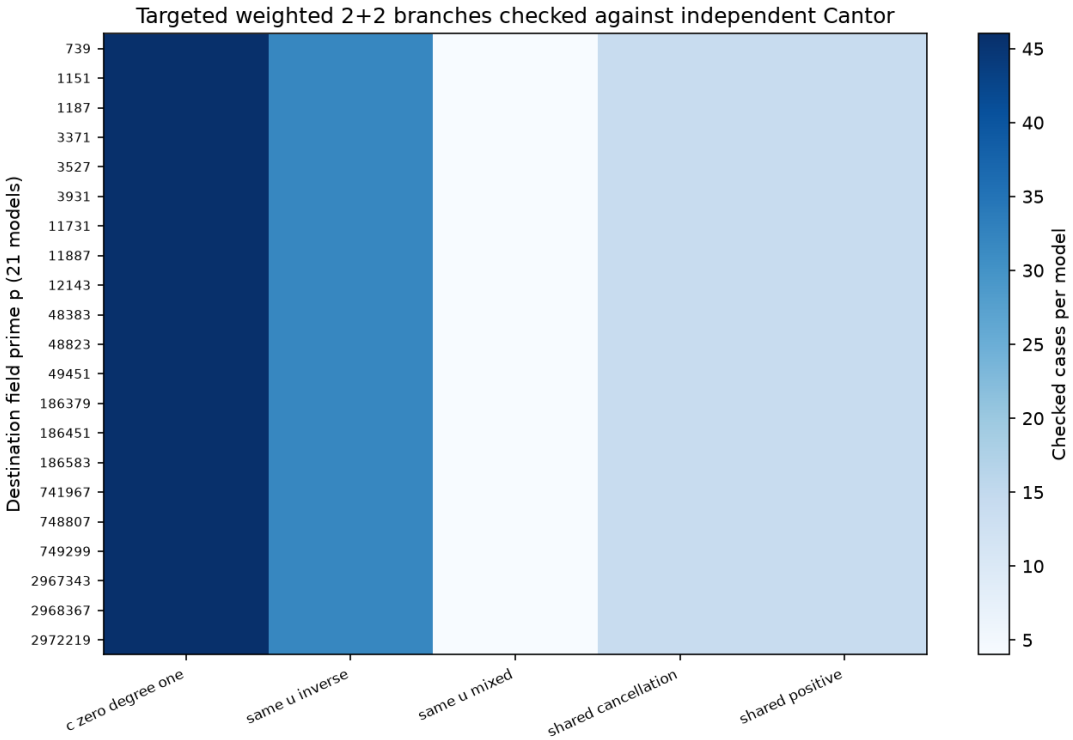
Values above are millions of source-addition equivalents for 180 paired 40-bit targets at the representative price. Setup is construction, transfer, and walk initialization, or rho initialization. Full per-size and pooled phase counts are in the CSV.





Arithmetic gate. Raw branch records cover all 21 normalized models with non-unit Z, W checks and targeted inverse, coincident-support, degree-drop, and doubling cases. The separate $p = 739$ regression preserves the exact input and both N_6 variants. These are finite checks, not a proof of universal branch correctness.

Later proof audit. Another 1,224,000 weighted additions on 42 models matched Cantor with no unresolved branches. Eight exact integer-polynomial certificates and written divisor arguments cover the reachable 1+1, 1+2, and 2+2 branches on smooth monic quintics without an x^4 term over odd-characteristic fields, for valid reduced inputs and nonzero weights. The certificate formulas are manually matched to runtime code; no mechanized theorem or independent mathematical peer review is recorded. See `PROOF_AUDIT.md`.



Review files. PROTOCOL.md defines the cohort and accounting. results/ contains raw per-target JSONL, source-replay and price summaries, both 36-cell grids, and the regression JSON. DATA_DICTIONARY.md defines every standardized field. VALIDATION_NOTES.md lists checks and remaining gaps. The SVG and PNG share the same recorded-values source.